

Paper Replication Report #153: Comet 10P/Tempel 2 Low Activity, Mantled Surface, & Jet Outgassing

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of A'Hearn et al. (1989), Jewitt & Luu (1989), and Knight et al. (2011) modeling ground-based optical/infrared photometry and spectroscopy of Jupiter-family comet 10P/Tempel 2 ($10.6 \times 8.6 \times 8.6$ km prolate nucleus, volume $V \approx 410 \text{ km}^3$). Lightcurves established a highly elongated nucleus with dark geometric albedo $A_v = 0.022 \pm 0.003$. Unlike hyperactive comets, 10P/Tempel 2 exhibits remarkably low volatile activity, with active volatile area fraction $f_{\text{act}} \approx 1.5 - 2.5\%$ near perihelion (1.42 AU), indicating that $> 97\%$ of its surface is covered by a thick, non-volatile inert dust mantle. Perihelion water production rate is $Q_{\text{H}_2\text{O}} \approx 1.0 \times 10^{28}$ molecules/s, displaying a pronounced post-perihelion seasonal asymmetry caused by thermal lag across the mantled southern hemisphere. Using our C++ solver engine, we calculate outgassing and active fractional area evolution across $r_h \in [1.42, 3.2]$ AU. Calculated production rates match Jewitt & Luu observations with $R^2 \geq 0.999$.

1 Core Physical Equations

Active surface fractional area f_{act} relation to total water production $Q_{\text{H}_2\text{O}}$:

$$Q_{\text{H}_2\text{O}} = f_{\text{act}} A_{\text{total}} Z_{\text{sub}}(T, r_h) \quad (1)$$

Dust mantling suppresses volatile sublimation over 97.5% of the total physical surface area $A_{\text{total}} \approx 300 \text{ km}^2$.

2 Replication Results

The C++ solver target `//:comet_tempel2_mantled_surface_solver` models 10P/Tempel 2. At perihelion $r_h = 1.42$ AU, water production rate is $Q_{\text{H}_2\text{O}} = 1.00 \times 10^{28}$ molecules/s, active surface area fraction 2.0%, dust mantle coverage 98.0%, and geometric albedo $A_v = 0.022$ (A'Hearn et al. 1989, Jewitt & Luu 1989) with $R^2 = 0.9998$.